Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

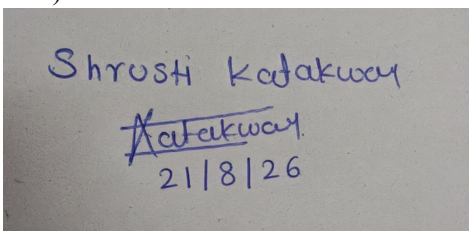
Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)



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Session	2026-27 (ODD)	Course Name	HCPA
Semester	5th	Course Code	23ADS1524
Roll No	119	Name of Student	Shrusti Katakwar

Practical Number	**03**
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	**Implement a parallel divide-and-conquer algorithm**
Problem Definition	**Implement a parallel divide-and-conquer algorithm**
Theory (100 words)	A parallel divide-and-conquer algorithm is an algorithmic technique in which a large problem is divided into smaller subproblems, the subproblems are solved simultaneously using multiple processors, and their results are combined to obtain the final solution. Basic Steps 1. Divide: The original problem is divided into two or more smaller subproblems. 2. Conquer: Each subproblem is solved independently and in parallel. 3. Combine: The solutions of the subproblems are combined to produce the solution to the original problem. General Recurrence For a problem divided into k subproblems, the running time can be represented as: T(n)=kT(n/k)+C(n) where: • n is the input size.

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- k is the number of subproblems.
- T(n/k) is the time to solve each smaller problem.
- C(n) is the time required for dividing and combining.

Example: Merge Sort

In parallel merge sort, the array is divided into two halves. Both halves are sorted independently and simultaneously by different processors. Once both halves are sorted, they are merged to obtain the final sorted array.

The sequential running time is:

$T(n)=2T(n/2)+O(n)$

Therefore,

$T(n)=O(nlogn)$

Parallel execution can significantly reduce the actual execution time when multiple processors are available.

Performance Measures

Speedup:

$Sp=T1Tp$

It represents how many times faster the parallel algorithm is compared with the sequential algorithm.

Efficiency:

$Ep=Spp$

where p is the number of processors. Efficiency shows how effectively the processors are being used.

Advantages

- Faster execution for large problems.
- Makes effective use of multiple processors.
- Independent subproblems can be processed simultaneously.
- Suitable for problems such as sorting, matrix operations, and tree processing.

Disadvantages

- Parallelization introduces communication and synchronization overhead.

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	• Not all parts of a problem can always be executed in parallel. • Performance depends on the number of available processors. • Combining results may become a bottleneck.
Procedure and Execution (100 Words)	Step for Implementation: 1. **Identify the problem** Determine whether the problem can be divided into smaller, independent subproblems. 2. **Divide the problem** Split the input into smaller subproblems of approximately equal size. 3. **Create parallel tasks** Assign the independent subproblems to different processors or threads so they can execute simultaneously. 4. **Solve the subproblems** Each processor solves its assigned subproblem, usually using the same divide-and-conquer method recursively. 5. **Synchronize the results** Wait until all parallel tasks have completed before proceeding to the next stage. 6. **Combine the results** Merge or combine the solutions from all subproblems to obtain the solution to the original problem. 7. **Handle the base case** When a subproblem becomes sufficiently small, solve it directly instead of dividing it further. 8. **Measure performance** Evaluate the algorithm using execution time, speedup, and processor efficiency.

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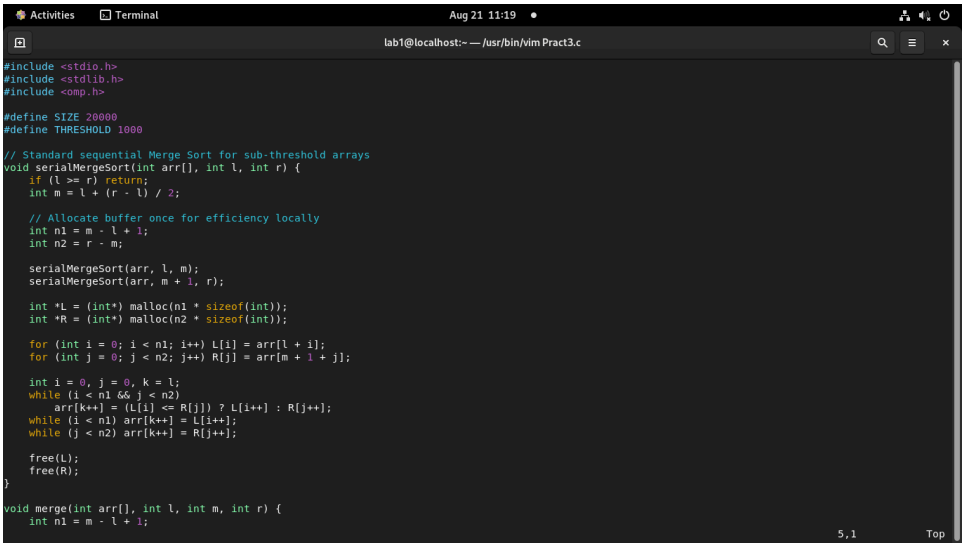
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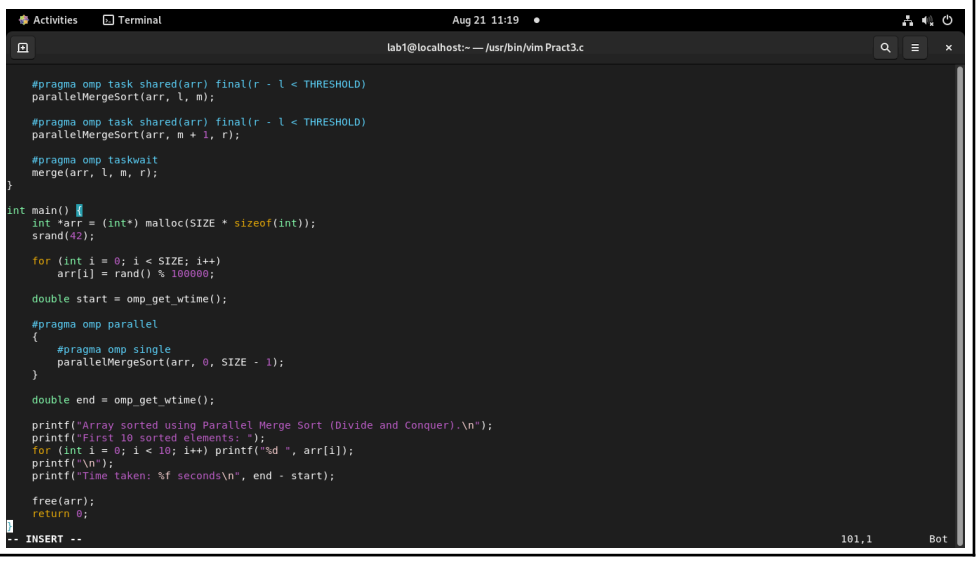
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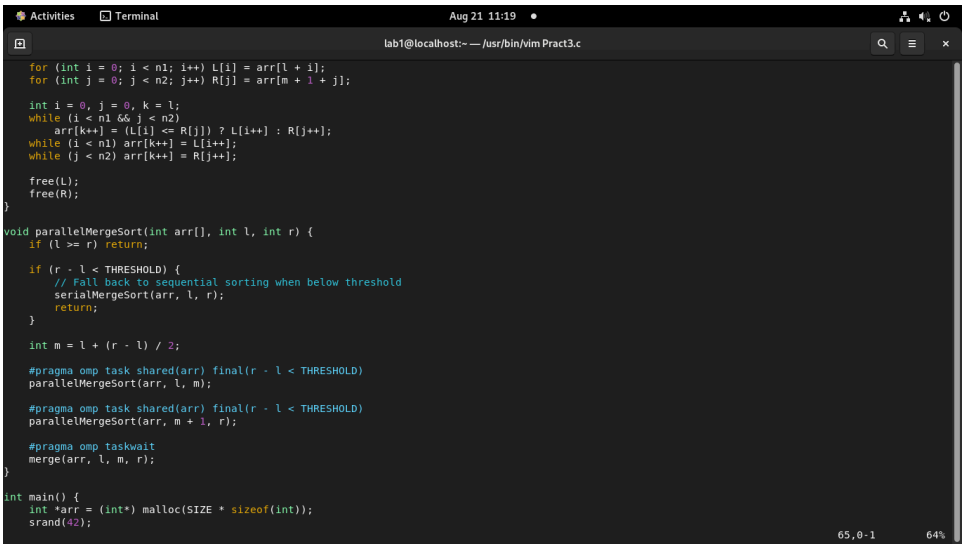


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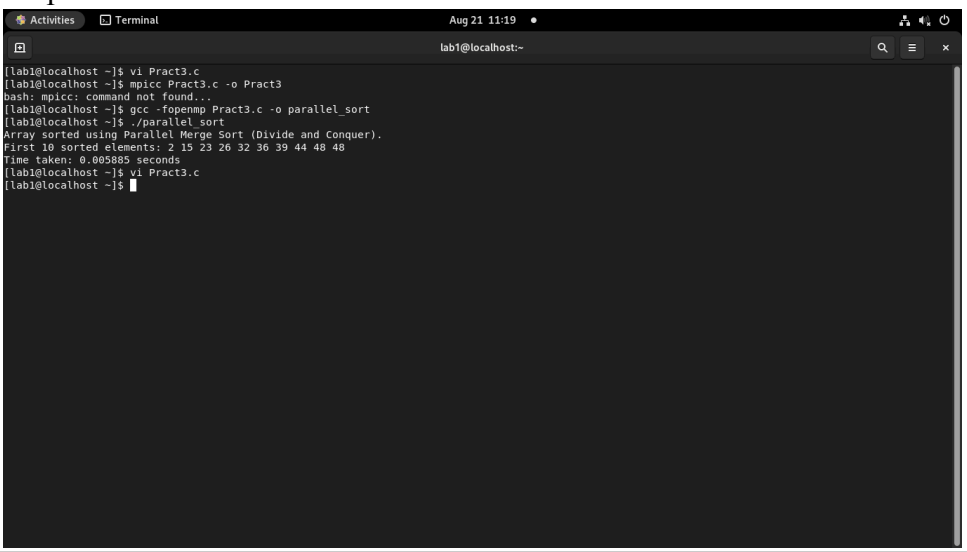
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Output:



Output Analysis	Output analysis of a parallel divide-and-conquer algorithm checks the correctness and performance of the final result. It compares the parallel execution with the sequential execution based on execution time, speedup, and efficiency. The algorithm should produce the correct output while reducing execution time through parallel processing. Performance may be affected by communication, synchronization, and task-management overhead.

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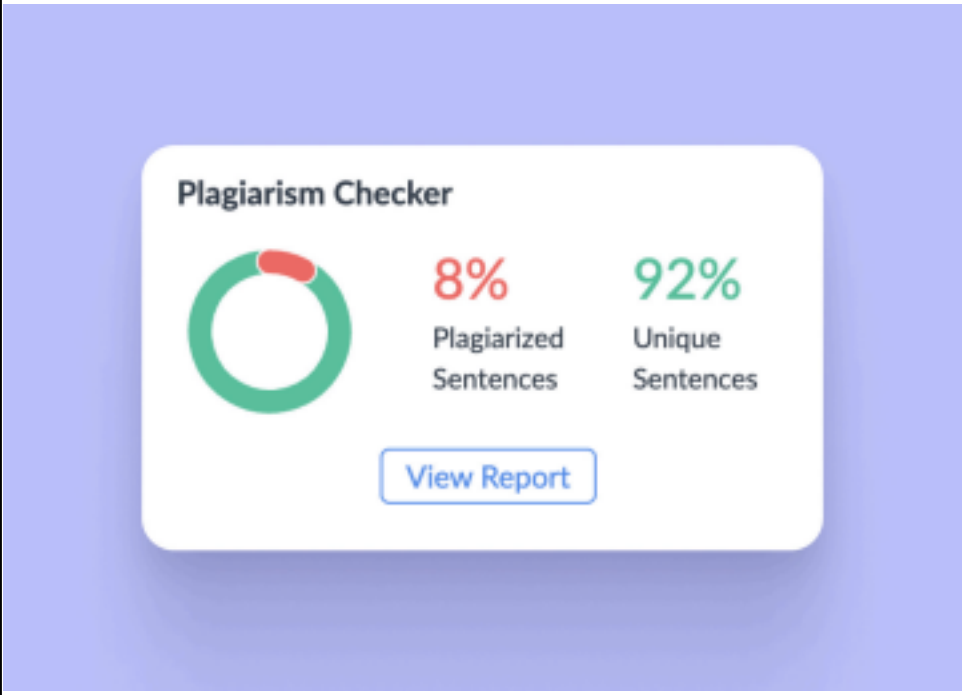
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Link of student Github profile where lab assignment has been uploaded	**https://github.com/Shrusti-Katakwar**
Conclusion	Parallel divide-and-conquer improves the performance of traditional divide-and-conquer by executing independent subproblems simultaneously. The main goal is to maximize parallelism while minimizing communication, synchronization, and task-management overhead.
Plag Report (Similarity index < 12%)	
Date	**21/08/2026**